pFedMLKD: A Novel Framework for Personalized Federated Learning via Multilevel Distillation

Yuankun Xia, Hui Wang, *Member, IEEE*, Di Yin, Yiteng Zhang, and Qian Huang

***Abstract*—With the evolution of edge intelligence technologies, federated learning has become a prominent decentralized learning framework, facilitating joint model training among numerous devices without compromising data privacy. However, federated learning confronts several challenges in practical applications, including data heterogeneity and edge device instability. To tackle these issues, this article introduces pFedMLKD, an innovative personalized federated learning framework that incorporates hierarchical knowledge distillation and leverages historical global model insights to enhance local model training, thereby reducing the effects of data heterogeneity. This method enhances learning efficiency and improves the generalization of models within federated learning systems. We establish theoretical guarantees for pFedMLKD's optimization under nonconvex settings through rigorous convergence analysis. Empirical validation encompasses three benchmark datasets (CIFAR-10, CIFAR-100, and EMNIST) under strict nonindependent and identically distributed data partitioning protocols. The results demonstrate that our algorithm outperforms current methods in federated learning scenarios.**

***Index Terms*—Data heterogeneity, edge intelligence, knowledge distillation (KD), personalized federated learning (PFL).**

I. Introduction

RECENTLY, driven by the fast advancement in mobile networking, large-scale data analytics, and Internet of Things (IoT) innovations, the massive quantity of data has been generated to provide a wealth of resources for artificial intelligence applications [1], [2]. However, traditional centralized machine learning methods often rely on storing data centrally on servers for training, posing significant risks to data privacy and security [3], [4], [5]. Consequently, protecting user privacy and ensuring data security have become critical issues that must be addressed.

Federated learning, an innovative distributed machine learning framework, has been proposed to address the aforementioned challenges [6], [7]. At its core, federated learning decentralizes the training process by distributing it among multiple edge devices, allowing each device to conduct data processing and model training on-site. After local training, by transmitting the model updates to a central server where aggregation occurs, this methodology not only ensures robust protection of data privacy but also significantly lowers the volume of data transmission required during the process. FedAvg [8], a foundational federated learning methodology, coordinates decentralized local training with local aggregation to achieve collaborative model optimization. However, optimization of communication efficiency is achieved primarily under the assumption of independent and identically distributed (IID) data between clients.

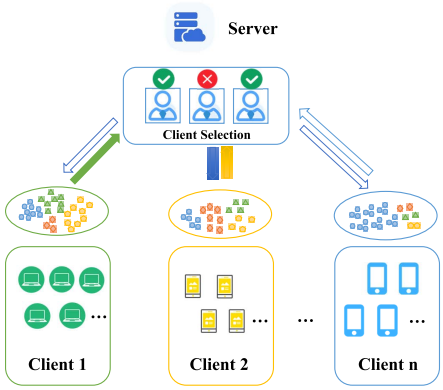

Fig. 1. Heterogeneous distribution of data on different clients. In the edge environment, the data distribution among different devices is in a non-IID form, resulting in heterogeneity.

In the real world, two serious challenges are faced: 1) the data heterogeneity and 2) the instability of edge devices. The heterogeneous nature of data, characterized by its non-independent and identically distributed (non-IID) structure, substantially affects the performance [9], [10], [11], as depicted in Fig. 1. Due to the different data distribution and optimization objectives, a unified minimization strategy may not be feasible. When clients prioritize their local data patterns during training, the aggregated global model often exhibits compromised effectiveness for the common task.

Personalized federated learning (PFL) frameworks are introduced to mitigate non-IID challenges through client-specific adaptation strategies during global model optimization [12], [13]. This approach enhances the model's performance for individual users and protects data privacy. The motivation for PFL comes from practical applications. Data generated by different clients often varies significantly in size and distribution, which is relevant to the global model. Due to these inherent disparities, a singular global model may not provide accurate predictions or optimizations for individual clients [14]. The challenge of performance degradation in PFL, particularly pronounced under conditions of data heterogeneity, necessitates

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the development of algorithms capable of generating client-specific models that preserve distinctive data patterns while fostering synergistic collaboration through adaptive parameter-sharing mechanisms.

A key challenge in PFL is that the global model cannot frequently offer specific or tailored guidance for local models. Knowledge distillation (KD) [15] presents a solution by driving a deliberate transfer of personalized knowledge between models, which is essential for optimal performance. Existing PFL methods often rely on single-level distillation. For example, FedGKD [16] uses global logit alignment to mitigate heterogeneity but ignores instance and class-level structures. FedGen [17] employs a generator to synthesize data-free knowledge, but its effectiveness depends on the generator's ability to mimic diverse client distributions, which is challenging under severe heterogeneity.

In the domain of PFL, the challenge lies in training client-specific models while maintaining global collaboration. A study on designing an efficient personalized update mechanism that ensures global coordination plays a key role [18]. As demonstrated by pFedSD [19], the global model often performs less effectively than local models on client datasets. Initializing global and local models at the start of a federated learning round can lead to the loss of previous knowledge, resulting in a decline in performance. This problem becomes more pronounced in environments with significant data heterogeneity. To tackle this challenge, FedKD [20] leverages KD to handle data heterogeneity, but the limitations in the effectiveness of knowledge transfer within traditional KD hinder its ability to effectively balance the performance of local and global models [21], [22].

In this article, we introduce a novel framework, pFedMLKD, which addresses the limitations of existing methods (e.g., FedGKD [16] and FedGen [17]) by integrating multilevel KD and global ensemble teacher models. Unlike FedGKD, which relies on single-level global distillation, pFedMLKD performs distillation at the instance-, batch-, and class-levels to capture fine-grained knowledge (e.g., instance-specific predictions, batch-wise data correlations, and class-level semantic relationships). Additionally, instead of using large pretrained models as teachers (as in FedGKD), we employ an ensemble of historical student models as the teacher, which is lightweight and stable for edge devices. This design avoids the computational overload of edge devices and enhances the adaptability to data heterogeneity.

Focusing on logit distillation, our approach does not require access to features from intermediate layers. In particular, we introduce a multilevel logit distillation approach, which enhances the extraction of information from logit outputs. Through a multilevel distillation framework, we reduce the prediction differences between the teacher and student models across instance, batch, and class levels. It enables the student model to learn instance-specific predictions, batch-level relationships, and class-level semantic correlations from the teacher model. To improve the alignment between the student and teacher models, particularly with constrained computational capabilities on edge devices, we use the aggregated ensemble of student models as the teacher model for the next round of distillation, effectively reducing the load on edge devices. Through the process of KD, the proposed framework demonstrates robust capability in mitigating non-IID data challenges inherent to cross-device federated environments. Three key innovations distinguish our approach.

1) We introduce a novel PFL framework combining multilevel logit and global distillation. The framework handles PFL from three dimensions, striking an improved balance between personalization and generalization while efficiently accounting for the computational capacity of edge devices. Besides, we theoretically analyze the pFedMLKD on convergence performance.
2) We use a global ensemble approach to select the teacher model, which avoids the instability of large-scale pretrained models on edge devices. Moreover, this method is more beneficial for the student model to learn from the teacher model, speeding up the convergence of the model.
3) Our proposed method is rigorously evaluated against leading PFL techniques across multiple real-world data environments and experimental configurations. The experimental results reveal that our method achieves outstanding performance, robustness, and effectiveness in training personalized models.

II. Related Work

A. Data Heterogeneity

In the research of FL, the issue of data heterogeneity presents a notable challenge, as it pertains to the non-IID characteristics of data distributions among participating clients [23], [24]. This issue leads to a substantial decline in model accuracy, impacting the model's generalization capability on the global dataset [25], [26], [27]. Researchers have developed a variety of techniques to tackle the challenges stemming from non-IID datasets. However, due to the experiments of these algorithms being based on different non-IID scenarios, it is challenging to systematically and intuitively comprehend their advantages and weaknesses. At the data level, efforts are directed toward reducing local data heterogeneity or improving privacy-preserving measures, including techniques like data augmentation and anonymization [28]. For instance, FedGen integrates local model knowledge by learning a lightweight generator and broadcasts it to clients in a data-free manner to regularize local training.

Furthermore, FedDF [29] proposes an ensemble distillation-based model fusion approach, in which the server utilizes unlabeled proxy data to conduct KD on local models, facilitating the transfer of knowledge from local to global models. At the model level, researchers concentrate on model design manipulations, such as sharing local structures and model optimization. Server-level approaches involve the server in activities like choosing clients or organizing them into clusters. For instance, FedProx [30] tackles data heterogeneity by introducing a proximal term into the optimization process, which restricts local model updates to remain closeness to

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the global model. Furthermore, FedFed [31] tackles data heterogeneity in federated scenarios through feature distillation, pioneering the extraction and sharing of partial features in data to significantly improve federated learning performance and convergence speed in heterogeneous data scenarios.

B. Knowledge Distillation

KD is widely utilized in deep learning and was originally introduced to enable knowledge transfer from a complex teacher model to a simple student model. It helps the student model deliver improved performance even with restricted computational and storage capabilities [32], [33], [34]. As deep learning models continue to advance, researchers have proposed numerous innovative KD methods, many of which have been refined to address specific application scenarios and needs. The fundamental KD method, proposed by Hinton [15], is based on using the soft labels generated by the teacher model to direct the learning of the student model. FitNets [35] introduced a novel KD approach that learns from the intermediate-layer outputs of the teacher model as targets to aid in the training of the student model. Unlike traditional KD methods that focus solely on the final output, FitNets concentrates on the intermediate representations of the teacher model, aiding the student model in learning more fine-grained feature representations and thereby enhancing its learning capabilities [35].

The self-ensembling method proposes an improvement to KD by integrating the student model itself, avoiding the need for an independent teaching model. Instead, the student model generates soft labels through a maintained historical model during training. TAKD [36] method enhances the distillation process by incorporating task-relevant knowledge. This method employs knowledge from various subtasks within the task to support the learning of the student model. Conditional KD (CKD) [37] introduces a conditional learning strategy within the KD process. By incorporating conditional variables, the strategy can be dynamically adjusted based on different characteristics of the input data. This method, through learning conditional distillation targets, can better capture complex relationships between inputs and outputs, improving the student model's performance on complex tasks. While these advancements in enhanced KD show promise in improving performance, they may lead to higher communication, storage, and computational costs in FL systems. This is due to the need to exchange more sophisticated information among models, which could result in increased resource consumption.

In FL, FedGKD [16] employs global KD by transmitting a pretrained teacher model to clients, but its single-level distillation (focusing only on global logit alignment) fails to capture fine-grained data heterogeneity. FedGen [17] uses a generator to distill knowledge without data, but it lacks explicit modeling of hierarchical data structures. In contrast, pFedMLKD introduces multilevel distillation (instance, batch, and class-level) to address diverse heterogeneity challenges. Instance-level distillation mimics teacher predictions for individual samples, addressing local data distribution shifts. Batch-level distillation aligns sample correlations within batches, capturing global data structure. Class-level distillation models semantic relationships between classes, enhancing generalization. Moreover, unlike FedGKD's reliance on large pretrained teachers, pFedMLKD uses an ensemble of historical student models as the teacher, which is computationally lightweight and avoids edge device instability.

III. Proposed pFedMLKD

This section offers an overview of the background knowledge and formally defines the problem statement.

A. Preliminaries

FL: Federated learning represents a decentralized architecture within the domain of machine learning, designed to allow numerous clients to collectively participate in the training process of a shared model, ensuring data privacy without sharing actual data. Consider a federated learning setup with N clients, denoted as $C_1, C_2, \ldots, C_N$, where each client C_i holds its own local dataset D_i. The global model is represented by the parameters w. Each client C_i employs their local data D_i and the current global model w_t to train their local model. FL aims to address the optimization problem given below [1]

$$F_i(w_i, x) = \frac{1}{|D_i|} \sum_{x \in D_i} \mathcal{L}_i\left(w_i^t; x\right) \tag{1}$$

where $F_i(w_i, x)$ represents the objective function of the FL and $\mathcal{L}_i(w_i^t; x)$ represents the loss function computed by client C_i using the sample (x, y). Upon completion of local training, the server collects updated models from all clients for aggregation:

$$\mathbf{w}_{t+1} = \sum_{i=1}^{N} \frac{|D_i|}{\sum_{i=1}^{N} |D_i|} \mathbf{w}_i^t \tag{2}$$

KD: KD is a model compression technique that facilitates the transfer of expertise from a larger teacher model to a smaller, more efficient student model. The fundamental concept of KD lies in leveraging the probability distribution outputs from the teacher model to guide and supervise the training process of the student model. A widely used approach in data-free KD leverages generative adversarial networks (GANs) [17], in which a generator is trained to create artificial training data as a substitute for the original dataset, enabling the KD process without the need for actual data. We begin with the vanilla KD method proposed in [15].

The logit output for an individual input is represented by $Z \in \mathbb{R}^C$, where C denotes the number of distinct data categories, and the class probabilities are given by

$$P_j = \frac{e^{Z_j/T}}{\sum_{k=1}^{C} e^{Z_k/T}} \tag{3}$$

where Z_j represents the logit output of an individual input and P_j represents the predicted probability of the jth class. We calculate the Softmax values with T, which helps mitigate the over-confidence problem in neural networks [38].

In KD, when the temperature is equal to one, it degenerates into the softmax function. Hence, in our experiments, we set the temperature value to be greater than one. The goal of

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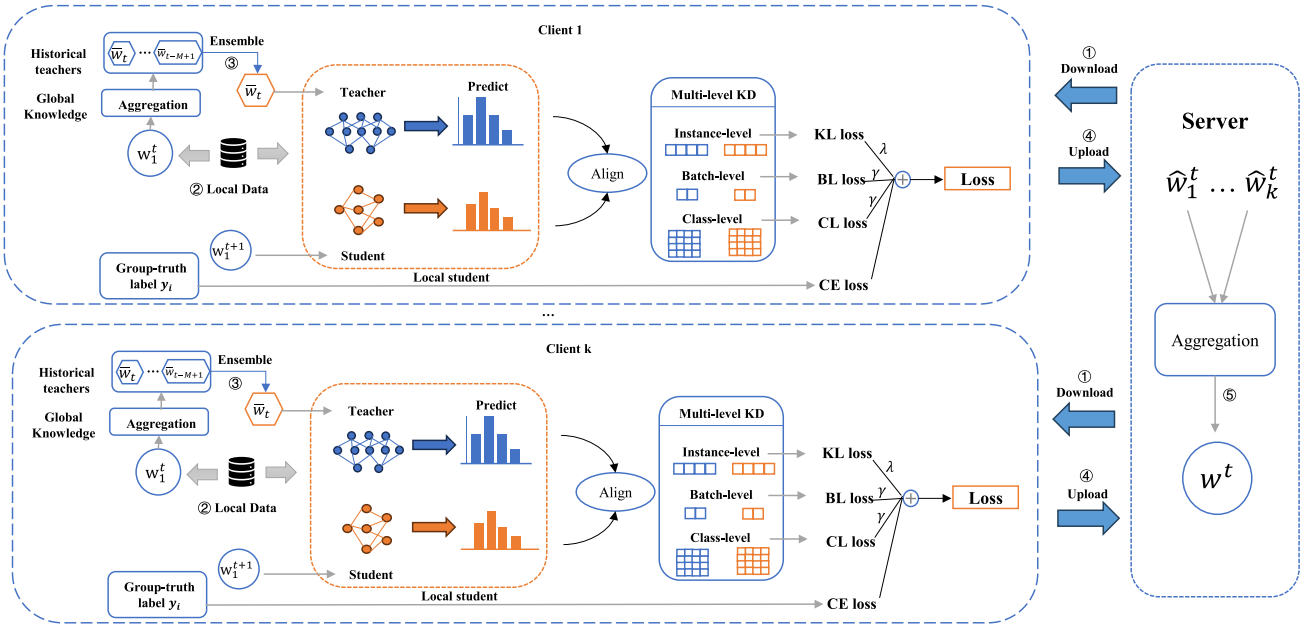
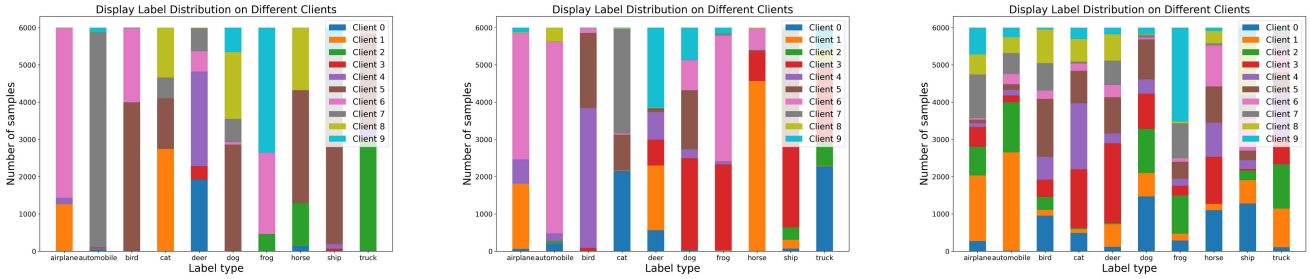


Fig. 2. Illustration of our pFedMLKD workflow. This workflow consists of five steps: 1) the server chooses a group of clients and sends the global model w_t to them; 2) each active client k initializes its local model $w_{t,k}$ based on the received global model, and then carries out local training using multilevel KD; 3) after training, the client combines the updated local model and saves it as the teacher model for the next round; 4) the client sends the updated local model back to the server; and 5) the server collects all the local models and combines them to form a new global model w_{t+1}.



(a) (b) (c)

Fig. 3. Visualization of statistical heterogeneity across clients in the CIFAR-10 dataset with varying α values. The bar length represents the number of training samples for each client's available labels. (a) $\alpha = 0.1$. (b) $\alpha = 0.2$. (c) $\alpha = 1.0$.

knowledge transfer lies in bridging the gap between larger teacher models and smaller student models through optimized learning strategies. To accomplish this, cross-entropy (CE) loss serves as a critical metric to quantify discrepancies between predicted outputs of the student model and corresponding groundtruth labels

$$L_{\mathrm{CE}} = \mathrm{CE}(F_i(w_i, x), y) = \frac{1}{|D_i|} \sum_{(x_k, y_k) \in D_i} l(F_i(w_i, x_k), y_k) \tag{4}$$

where $F_i(w_i, x)$ represents the predicted values of the student model without temperature scaling and $l(F_i(w_i, x_k), y_k)$ represents the CE value between the predictions of the student model and the ground-truth values.

Additionally, through output rescaling, the KD approach accomplishes distillation by reducing the Kullback–Leibler (KL) divergence between the predictions of the teacher and student models, expressed as

$$L_{\mathrm{KD}} = \mathrm{KL}(p_{\mathrm{tea}} || p_{\mathrm{stu}}) = \sum_{j=1}^{C} p_j^{\mathrm{tea}} \log\left(\frac{p_j^{\mathrm{tea}}}{p_j^{\mathrm{stu}}}\right) \tag{5}$$

where p_j^{tea} and p_j^{tea} denote the probability values for the jth class of the teacher and student model outputs, respectively.

B. Problem Statement

This section provides an overview of pFedMLKD, as shown in Fig. 2 and Algorithm 1. The main goal is to tackle data heterogeneity, where client data distributions vary significantly (e.g., non-IID label distributions in Fig. 3), making a single global model insufficient for personalized tasks. pFedMLKD addresses this by multilevel distillation [39] and global ensemble distillation [16], which jointly capture fine-grained and global knowledge to align local models with diverse data patterns.

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Algorithm 1 Framework of pFedMLKD

Input: total number of clients is denoted by K, total communication rounds T, fraction of participation clients e, local epochs E_{local}, data batch B_{batch}, learning rate η
Output: global model w^t
1: **Server Execution:**
2: Initial global model w_0
3: **for** each round of communication, where $t = 1, ..., T$ **do**
4: $K^t \leftarrow$ select a random set of $C \cdot K$ clients
5: Transmit the global model w^t and ensemble model $\bar{w}^t$ to the chosen clients
6: Store the global model w_t
7: **for** each client $k \in K$, in parallel **do**
8: $w_{t+1}^k \leftarrow$ ClientUpdate(k,w_t)
9: $w_{t+1}^k \leftarrow \sum_{k=1}^{K} \frac{n_k}{n} w_{t+1}^k$
10: **end**
11:
12: **ClientUpdate**(k, w_t)
13: **for** $w \leftarrow w_t$
14: **for** each local epoch $i = 1, ..., E$ **do**
15: **for** each batch $b \in B$ **do**
16: $w \leftarrow w - \eta \cdot \nabla L(D_i, w_i)$ //Update the local model via minimizing $L(D_i, w_i)$.
17: **end**
18: **end**
19: **end**
20: Ensemble the models $w_1, ..., w_t$ as the teacher model
21: **Return** w back to server

We deploy pFedMLKD within a client-server framework. In every communication round, the server randomly selects a group of clients to participate and shares the aggregated global knowledge with them (Step 1). Clients leverage the global model obtained to direct their respective localized training initiatives (Step 2). We primarily conduct local training through multilevel distillation and a prediction enhancement mechanism [39]. Following that, clients transmit their models to the server, which unites them into a global model. This global model acts as a teacher to direct the training of the student models (Step 3). The clients subsequently send their updated local models to the server (Step 4). The server then combines these models to generate an updated global model (Step 5).

Initially, we employ a multilevel distillation method that uses a model prediction enhancement mechanism to supervise local training. Thereafter, we form an ensemble of past global models as a teacher model to guide the training process.

C. Multilevel KD

In this section, we introduce our approach to multilevel KD. Unlike traditional KD, which operates on individual data points, we utilize batch data. Hence our output is denoted as $Z \in \mathbb{R}^{B_{\text{batch}} \times C}$, where B_{batch} signifies the batch size and C represents the number of classes. Besides, we select M temperatures for model predictions to achieve enhanced distillation efficiency

$$P_{i,j,m} = \frac{e^{Z_{i,j}/T_m}}{\sum_{k=1}^{C} e^{Z_{i,k}/T_m}} \tag{6}$$

where $P_{i,j,m}$ represents the softened probability for the client i and class j, calculated on a grid of m temperatures, thus enriching the distillation process. Furthermore, we employ multilevel KD for model training. The instance-level distillation addresses local data distribution bias by forcing the student model to mimic the teacher's predictions for individual samples. For non-IID data, where clients may have skewed class distributions (e.g., one client dominated by "airplane" samples in CIFAR-10), L_I ensures the student learns instance-specific features, reducing misclassification caused by local data shifts

$$\begin{aligned} L_I &= \frac{1}{|D_i|} \sum_{(x_k, y_k) \in D_i} \sum_{m=1}^{M} \mathrm{KL}\left(p_{x_k,m}^{\text{tea}} \| p_{x_k,m}^{\text{stu}}\right) \\ &= \frac{1}{|D_i|} \sum_{(x_k, y_k) \in D_i} \sum_{m=1}^{M} \sum_{j=1}^{C} p_{x_k,j,m}^{\text{tea}} \log \frac{p_{x_k,j,m}^{\text{tea}}}{p_{x_k,j,m}^{\text{stu}}} \end{aligned} \tag{7}$$

where L_I denotes the loss augmented by predictions from multiple temperatures and M represents the number of temperatures. This KD method aims to ensure that the student model closely imitates the teacher model, constituting the most critical component in the KD process.

In addition to using the classic instance-level distillation, we also incorporate batch- and class-level logit distillation to address the shortcomings of previous single-direction approaches. We perform prediction by comparing the input correlations using the Gram matrix and quantifying it using logit predictions, as detailed below

$$B_{ab}^m = \sum_{j=1}^{C} p_{a,j,m} \cdot p_{b,j,m}, C_{ab}^m = \sum_{i=1}^{C} p_{i,a,m} \cdot p_{i,b,m} \tag{8}$$

where B_{ab}^m and C_{ab}^m represent the similarity between the ath and bth classes at the batch- and class-level respectively, thus quantifying the relationship between the two categories.

The batch-level distillation leverages the Gram matrix to align intrabatch sample correlations, capturing global data structures that are disrupted in heterogeneous environments. For example, in non-IID scenarios where clients have divergent class distributions, L_B ensures the student model learns consistent sample relationships (e.g., "car" and "truck" are more similar than "car" and "bird"), improving generalization across clients.

The class-level distillation models interclass semantic relationships (e.g., "dog" versus "cat" in CIFAR-100), which are critical for non-IID data where clients may lack certain classes. By aligning class similarity matrices, L_C enhances the student's ability to generalize to unseen class combinations, addressing the "long-tail" heterogeneity problem

$$L_B = \frac{1}{B_{\text{batch}}} \sum_{m=1}^{M} ||B_{\text{tea}}^m - B_{\text{stu}}^m||_2^2 \tag{9}$$

$$L_C = \frac{1}{C} \sum_{m=1}^{M} ||C_{\text{tea}}^m - C_{\text{stu}}^m||_2^2 \tag{10}$$

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where L_B and L_C respectively denote the loss values at the batch- and class-level distillation. B^m_{tea}, B^m_{stu} and C^m_{tea}, C^m_{stu} are the predictions of the teacher and student models at temperature m at the batch- and class-level distillation respectively.

Now, we have established a multilevel KD framework, encompassing instance-, batch-, and class-level distillation. Our total loss function can be expressed as

$$\begin{aligned} L(D_i, w_i) = &\frac{1}{|D_i|}\sum_{(x_k,y_k)\in D_i} l(F_i(w_i, x_k), y_k) \\ &+ \lambda\frac{1}{|D_i|}\sum_{(x_k,y_k)\in D_i}\sum_{m=1}^{M}\sum_{j=1}^{C} p^{\text{tea}}_{x_k,j,m}\log\frac{p^{\text{tea}}_{x_k,j,m}}{p^{\text{stu}}_{x_k,j,m}} \\ &+ \gamma\frac{1}{B}\sum_{m=1}^{M}||B^m_{\text{tea}} - B^m_{\text{stu}}||_2^2 \\ &+ \gamma\frac{1}{C}\sum_{m=1}^{M}||C^m_{\text{tea}} - C^m_{\text{stu}}||_2^2 \end{aligned} \tag{11}$$

where λ and γ are the distillation coefficients for the global model. By integrating the four above components, our design aims to enhance the student's capacity to learn efficiently from the teacher's knowledge.

D. Global KD

In FL, due to edge devices' limited load capacity, using an oversized pretrained teacher model can lead to device instability. Therefore, this article integrates global student models to replace large-scale pretrained models as teacher models for KD. This ensures that these smaller models are compact enough to be executed directly on mobile devices, thus providing a promising means for on-device model execution. After completing local multilevel KD, clients transmit their training models to the central server. The server then archives and consolidates each client's model to be repurposed as the teacher model for subsequent training rounds [21].

To address the instability of large pretrained models on edge devices, we use an ensemble of historical student models as the teacher. This lightweight design avoids computational overload while leveraging the diversity of previous models to combat data heterogeneity. By aggregating models from multiple clients and rounds, w^{t+1}_{tea} captures robust global knowledge, reducing overfitting to the data distribution of any single client. This can be specifically represented as follows:

$$w^{t+1}_{\text{tea}} = w^T = \frac{1}{N}\sum_{i=1}^{N} w^t_i. \tag{12}$$

Here, w^t_i denotes the output of the student model after the tth round of distillation learning. N refers to the total number of students. w^{t+1}_{tea} and w^T represent the ensembled teacher model and global model parameter for the $(t+1)$th round respectively.

The ensemble uses simple averaging, requiring no complex operations, and distillation relies solely on logit outputs (not intermediate features), reducing computational overhead compared to methods like FitNets [35].

IV. Convergence Analysis

Considering the heterogeneous case of pFedMLKD, we prove the convergence of our algorithm under well-defined mathematical assumptions.

A. Assumptions

Assumption 1 (Lipschitz Continuity): The loss functions $L(D_i, w_i)$ are L-smooth, meaning that their gradients are Lipschitz continuous. For any w_1, w_2, we have the following expressions:

$$\begin{aligned} L(D_i, w_1) \le\ & L(D_i, w_2) \\ &+ \nabla L(D_i, w_2)^T(w_1 - w_2) + \frac{L}{2}\|w_1 - w_2\|_2^2. \end{aligned} \tag{13}$$

Assumption 2 (Unbiased Gradient): The gradient of each client is unbiased. Specifically, for any w^t_i, their expectation is equal to the global parameter w^T. This can be expressed as follows:

$$\mathbb{E}\big[\nabla L\big(D_i, w^t_i\big)\big] = \nabla L\big(w^T\big). \tag{14}$$

Assumption 3 (Bounded Gradient Norm): The gradient of each client is bounded. There exists a constant $P > 0$ with the following expression:

$$\|\nabla L\big(D_i, w^t_i\big)\| \le P \qquad \forall i, t. \tag{15}$$

Assumption 4 (Bounded KL Divergence): The KL divergence between the output distributions of the global central model and the local student models remains bounded. For any x_k, there is a bound in the KL divergence between $p^{\text{tea}}_{x_k,m}$ predicted by the teacher model and $p^{\text{stu}}_{x_k,m}$ predicted by the student model at temperature m

$$\text{KL}\big(p^{\text{tea}}_{x_k,m}\ ||\ p^{\text{stu}}_{x_k,m}\big)\ \le\ \text{KL}_{\max}. \tag{16}$$

B. Lemma

Lemma 1: Lemma 1 can be obtained from [40] with the following expression at the global t-th round in the Eth round of local training:

$$\begin{aligned} \mathbb{E}\Big[L\Big(D_i, w^{t+E}_i\Big)\Big] \le\ & L\big(D_i, w^t_i\big) - \eta E\mathbb{E}\Big[\|\nabla L\big(D_i, w^t_i\big)\|^2\Big] \\ &+ \frac{L\eta^2 E\big(P^2 + \sigma^2\big)}{2} \end{aligned} \tag{17}$$

where L represents the Lipschitz constant. σ^2 denotes the bound on the expected difference between the gradients of the local and global models. P represents the bound of the gradient paradigm of $L(D_i, w^t_i)$.

Lemma 2: Under Assumptions 1–4 and Lemma 1, the global loss function at round $t+1$ can be bounded as

$$\begin{aligned} \mathbb{E}\Big[F\Big(w^{t+1}\Big)\Big] \le\ & F\big(w^t\big) - \frac{\eta E}{n}\sum_{i=1}^{n}\mathbb{E}\Big[\|\nabla F_i\big(w^t_i\big)\|^2\Big] \\ &+ \frac{L\eta^2 E\sigma^2}{2} \end{aligned} \tag{18}$$

where $F(w^t)$ is the global loss function and $F_i(w^t_i)$ is the local loss function.

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Proof: From Lemma 1, we can get the following:

$$\mathbb{E}\left[F_i\left(w_i^{t+E}\right)\right] \leq F_i\left(w_i^t\right) - \eta E \mathbb{E}\left[\|\nabla F_i\left(w_i^t\right)\|^2\right] + \frac{L\eta^2 E\left(P^2+\sigma^2\right)}{2}. \tag{19}$$

We can denote the result of our loss function aggregated by clients at round $t+1$,

$$\mathbb{E}\left[F\left(w^{t+1}\right)\right] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}\left[F_i\left(w_i^{t+E}\right)\right]. \tag{20}$$

So we can perform a traversal client 1 to n operation for (19) and get the following result

$$\mathbb{E}\left[F\left(w^{t+1}\right)\right] \leq \frac{1}{n}\sum_{i=1}^{n}\left(F_i\left(w_i^t\right) - \eta E \mathbb{E}\left[\|\nabla F_i\left(w_i^t\right)\|^2\right]\right) + \frac{1}{n}\sum_{i=1}^{n}\left(\frac{L\eta^2 E\left(P^2+\sigma^2\right)}{2}\right). \tag{21}$$

According to the local model aggregation and $P > 0$, we can obtain

$$F\left(w^t\right) = \frac{1}{n}\sum_{i=1}^{n}F_i\left(w_i^t\right). \tag{22}$$

Therefore, the following inequality can be derived from the deflation relation (21):

$$\mathbb{E}\left[F\left(w^{t+1}\right)\right] \leq F\left(w^t\right) - \frac{\eta E}{n}\sum_{i=1}^{n}\mathbb{E}\left[\|\nabla F_i\left(w_i^t\right)\|^2\right] + \frac{L\eta^2 E\sigma^2}{2}. \tag{23}$$

■

C. Theorem

Theorem 1 (pFedMLKD Convergence): Under Assumptions 1–4, Lemma 1 and Lemma 2, our pFedMLKD has reached the convergence.

Proof: We can rearrange the Lemma 2 inequality to get

$$F\left(w^t\right) - \mathbb{E}\left[F\left(w^{t+1}\right)\right] \geq \frac{\eta E}{n}\sum_{i=1}^{n}\mathbb{E}\left[\|\nabla F_i\left(w_i^t\right)\|^2\right] - \frac{L\eta^2 E\sigma^2}{2} \tag{24}$$

and we accumulate the left side of the inequality from $t = 1,\ldots, E$

$$\sum_{t=1}^{E}\left(F\left(w^t\right) - \mathbb{E}\left[F\left(w^{t+1}\right)\right]\right) = F\left(w^1\right) - \mathbb{E}\left[F\left(w^{E+1}\right)\right]. \tag{25}$$

Next, we accumulate simultaneously for the left and right sides of inequality (24) from $i = 1,\ldots, E$:

$$F\left(w^1\right) - \mathbb{E}\left[F\left(w^{E+1}\right)\right] \geq \frac{\eta E}{n}\sum_{t=1}^{E}\sum_{i=1}^{n}\mathbb{E}\left[\|\nabla F_i\left(w_i^t\right)\|^2\right] - \frac{L\eta^2 E^2\sigma^2}{2}. \tag{26}$$

We can rearrange inequality (26) to get the following:

$$\frac{1}{E}\sum_{t=1}^{E}\sum_{i=1}^{n}\mathbb{E}\left[\|\nabla F_i\left(w_i^t\right)\|^2\right] \leq \frac{F\left(w^1\right) - \mathbb{E}\left[F\left(w^{E+1}\right)\right]}{\frac{\eta E}{n}E} + \frac{L\eta^2\sigma^2}{2}. \tag{27}$$

Assuming that the globally optimal loss function is F^*, we can get the following:

$$\frac{1}{E}\sum_{t=1}^{E}\sum_{i=1}^{n}\mathbb{E}\left[\|\nabla F_i\left(w_i^t\right)\|^2\right] \leq \frac{F\left(w^1\right) - F^*}{\frac{\eta E}{n}E} + \frac{L\eta^2\sigma^2}{2}. \tag{28}$$

Since the learning rate $\eta > 0$, this implies that the global loss function will converge as the number of local training rounds E increases indefinitely. ■

V. Experiments and Analysis

A. Experimental Setting

Datasets and Models: Our experimental framework conducts systematic benchmarking against canonical federated learning architectures in image recognition tasks, employing the ResNet-8 convolutional neural network as the structure of our model network [41]. We use three datasets (CIFAR-10, CIFAR-100 [42] and EMNIST) to conduct our experiments. We randomly allocate 80% of each client's data for model training and reserve the remaining 20% for validation.

Data Heterogeneity: Building on previous studies [29], [43], [44], we employ the Dirichlet distribution to model heterogeneous data. Our method employs the Dirichlet distribution $\mathbf{Dir}(\alpha)$, which features a tunable concentration parameter α. As α decreases, the degree of heterogeneity becomes more pronounced. As shown in Fig. 3, for the CIFAR-10 dataset, we perform data heterogeneity processing using α values of 0.1, 0.2, and 1.

Baselines: We assess our proposed model by comparing it with the latest FL models, including FedAvg [8], FedProx [30], FedGKD [16] and FedGen [17]. We provide a brief introduction to the aforementioned baseline models as follows.

1) *FedAvg:* FedAvg [8] stands as a foundational algorithm in federated learning, employing straightforward averaging for model aggregation. By utilizing this method, we can achieve optimized machine learning while safeguarding user confidentiality.
2) *FedProx:* FedProx [30] is an improved method based on FedAvg. By integrating a regularization term into the local training process, we improve the model's generalization capabilities while effectively addressing data heterogeneity issues.
3) *FedGKD:* FedGKD [16] significantly reduces model parameters by introducing global KD based on traditional KD, enhancing its suitability for deployment on edge devices.
4) *FedGen:* FedGen [17] uses a global generator to regulate local training during the learning process, effectively

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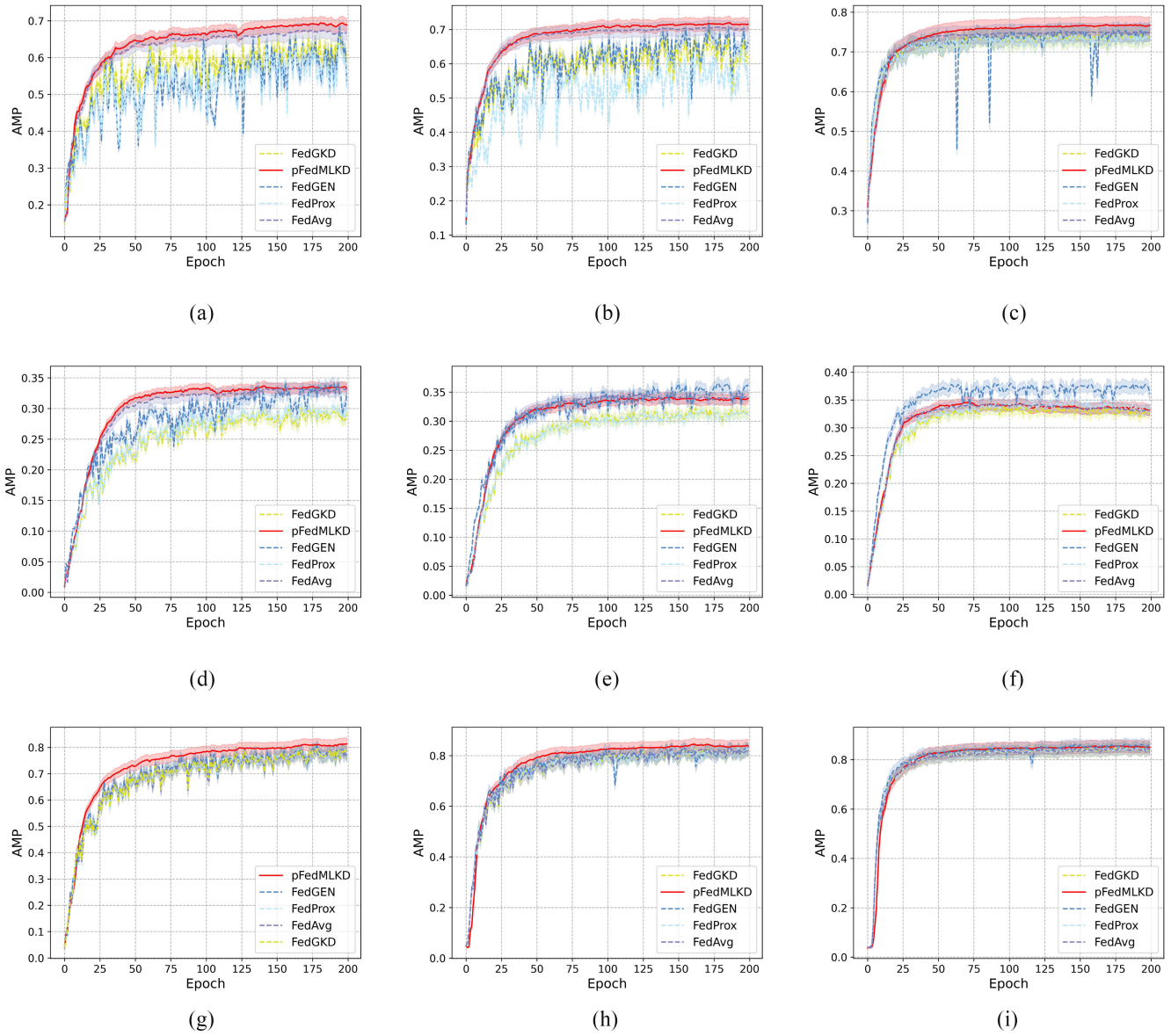

Fig. 4. Model Performance Evaluation. Performance on different datasets under various heterogeneity settings. (a)CIFAR-10 $\alpha = 0.1$. (b) CIFAR-10 $\alpha = 0.2$. (c) CIFAR-10 $\alpha = 1.0$. (d) CIFAR-100 $\alpha = 0.1$. (e) CIFAR-100 $\alpha = 0.2$. (f) CIFAR-100 $\alpha = 1.0$. (g) EMNIST $\alpha = 0.1$. (h) EMNIST $\alpha = 0.2$. (i) EMNIST $\alpha = 1.0$.

addressing the challenge of class imbalance in structured data.

Experimental Details: Both the proposed pFedMLKD and the comparison solutions are developed with PyTorch. Following the majority of previous research on FL, we conduct a simulation of the clients' computations on the server and subsequently assess the model's performance.

The optimizer used to train the local model is SGD. In our federated learning setup, the learning rate is set to 0.1, and we conduct a total of 200 communication rounds. The total number of clients is 20, with a selection rate C of 20%. The number of local update epochs E_{local} is 10, and the local batch size B_{batch} is 64. For pFedMLKD, we set the temperatures for the enhancement of model prediction to [2,3,4,5,6]. The λ in the loss function is set to 0.2 and the γ is set to 0.1.

B. Ablation Studies

We investigate the contribution of each component in our method, including the original CE loss, instance-level loss, batch-level loss, and class-level loss. Table I reveals the incremental value of each distillation level:

1) *Instance-level only* achieves 66.78% average model performance (AMP), confirming that mimicking individual samples is foundational but insufficient for heterogeneous data.
2) *Adding batch-level loss* improves accuracy to 67.58%, indicating that aligning batch-wise data correlations helps mitigate distribution shifts between clients.
3) *Full multilevel distillation* achieves the highest AMP (68.69%), showing that class-level semantic modeling (L_C) is critical for capturing long-range dependencies, especially under strong heterogeneity.

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TABLE I
ABLATION STUDY. THE EXPERIMENT IS CONDUCTED ON CIFAR-10 WITH $\alpha = 0.1$, USING AMP ACCURACY AS THE EVALUATION METRIC

CE Loss	Instance-level Loss	Batch-level Loss	Class-level Loss	AMP
✔	✗	✗	✗	66.78
✔	✔	✗	✗	66.93
✔	✔	✔	✗	67.58
✔	✔	✔	✔	68.69

TABLE II
OVERVIEW OF AMP ACCURACY TESTING UNDER DIFFERENT DATA SETTINGS FOR RESNET ACROSS VARIOUS DATASETS. **Bolded** NUMBERS INDICATE THE BEST RESULTS, AND UNDERLINED NUMBERS INDICATE THE SECOND-BEST RESULTS

Dataset	Dirichlet α	FedAvg	FedProx	FedGen	FedGKD	pFedMLKD(ours)
EMNIST	α = 0.1	$78.39_{\pm 0.34}$	$77.32_{\pm 0.96}$	$77.04_{\pm 0.75}$	$77.64_{\pm 0.31}$	$\mathbf{81.31}_{\pm 0.09}$
	α = 0.2	$82.09_{\pm 0.40}$	$82.13_{\pm 0.76}$	$81.77_{\pm 0.86}$	$82.01_{\pm 0.80}$	$\mathbf{83.81}_{\pm 0.20}$
	α = 1.0	$83.99_{\pm 0.75}$	$84.31_{\pm 0.24}$	$\mathbf{85.50}_{\pm 0.40}$	$84.03_{\pm 0.16}$	$85.02_{\pm 0.12}$
CIFAR-10	α = 0.1	$66.71_{\pm 0.43}$	$59.48_{\pm 1.26}$	$62.53_{\pm 1.41}$	$60.35_{\pm 1.14}$	$\mathbf{68.90}_{\pm 0.53}$
	α = 0.2	$70.48_{\pm 0.58}$	$60.99_{\pm 0.57}$	$66.38_{\pm 1.00}$	$61.68_{\pm 0.81}$	$\mathbf{71.59}_{\pm 0.19}$
	α = 1.0	$74.80_{\pm 0.19}$	$73.78_{\pm 0.32}$	$74.85_{\pm 0.39}$	$73.67_{\pm 0.22}$	$\mathbf{76.71}_{\pm 0.09}$
CIFAR-100	α = 0.1	$33.26_{\pm 0.11}$	$30.19_{\pm 0.72}$	$32.36_{\pm 1.09}$	$29.22_{\pm 0.80}$	$\mathbf{33.55}_{\pm 0.28}$
	α = 0.2	$34.35_{\pm 0.32}$	$33.52_{\pm 0.40}$	$\mathbf{36.31}_{\pm 0.66}$	$31.14_{\pm 0.85}$	$33.92_{\pm 0.14}$
	α = 1.0	$33.22_{\pm 0.25}$	$33.52_{\pm 0.40}$	$\mathbf{36.11}_{\pm 0.35}$	$32.67_{\pm 0.12}$	$33.63_{\pm 0.16}$

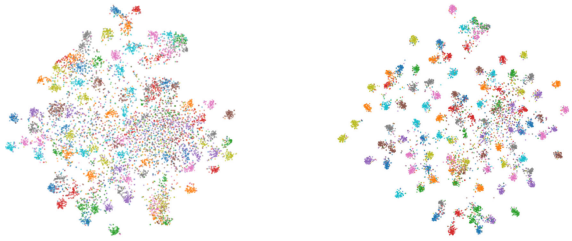
These results validate that batch- and class-level distillation are essential complements to instance-level KD, jointly addressing the hierarchical challenges of non-IID data.

C. Experimental Results

Model Performance Evaluation: The AMP [21] is used for model evaluation, which can adapt well to the federated learning environment. It calculates the average prediction of the selected clients as an evaluation metric. In this article, we also use this evaluation metric to assess our algorithm. We have studied the influence of varying Dirichlet distribution parameters α and federated learning configurations on the previously mentioned model performance, as shown in Table II and Fig. 4. PFedMLKD consistently outperforms baselines, particularly under high-data heterogeneity (low α values). For example, on CIFAR-10 with α =0.1 (strong non-IID), pFedMLKD achieves 68.90% accuracy, significantly higher than FedAvg (66.71%) and FedGKD (60.35%). This is because multilevel distillation captures fine-grained local structures (e.g., instance-specific features via L_I, batch-wise correlations via L_B) that single-level methods like FedGKD miss.

Notably, as α increases to 1.0, the performance gap between pFedMLKD and baselines narrows, e.g., on EMNIST, pFedMLKD's advantage over FedProx decreases from 6.49% ($\alpha = 0.1$) to 0.71% ($\alpha = 1.0$). This confirms that pFedMLKD is specifically optimized for heterogeneous environments, as its multilevel architecture becomes less critical when data distributions are balanced.

Model Validity Analysis: Taking CIFAR-100 with $\alpha = 0.1$ as an example, we compare the discriminability of deep features between the traditional KD model and pFedMLKD by using the t-SNE algorithm. As shown in Fig. 5, pFedMLKD's multilevel distillation results in more compact intraclass feature clusters and clearer interclass separations compared to FedKD [20] on CIFAR-100 (α =0.1). This demonstrates that capturing instance-, batch-, and class-level knowledge mitigates feature distribution divergence caused by non-IID data, enhancing model discriminability.



(a) (b)

Fig. 5. Model validity analysis. t-SNE visualizations of features of local models on the CIFAR-100 test dataset. (a) FedKD(left). (b) pFedMLKD(right).

Data Heterogeneity Analysis: To better explore the impact of heterogeneity on experimental results, we selected three algorithms, FedGen, FedGKD, and our pFedMLKD, to conduct experiments using five levels of heterogeneous data with $\alpha = 0.2, 0.4, 0.6, 0.8$, and 1.0, respectively, using CIFAR-10 as an example. The outcomes are presented in Table III and Fig. 6. It is evident that as the level of heterogeneity decreases, the model's predictive accuracy gradually increases. Compared with other algorithms, our pFedMLKD performs excellently under various heterogeneous data conditions.

Model Robustness Analysis: To evaluate pFedMLKD's robustness against client-level heterogeneity and system instability, we conduct experiments under varying client participation rates and client counts. We simulate scenarios where up to 30% of the selected clients drop out randomly

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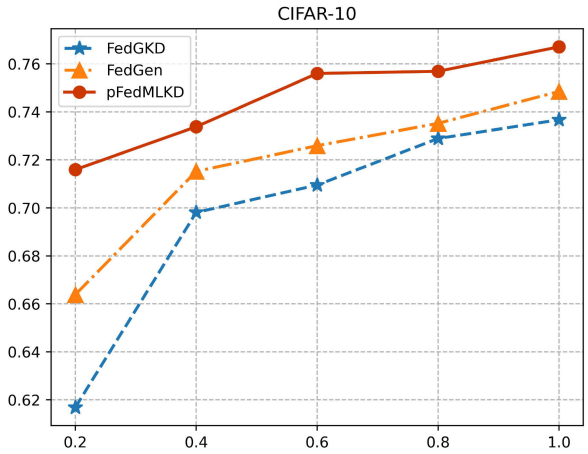


Fig. 6. Data heterogeneity analysis. Taking CIFAR-10 as an example, compare the results of FedGKD, FedGen, and FedMLKD models under different α heterogeneity rates.

TABLE III
DATA HETEROGENEITY ANALYSIS: COMPARING THE RESULTS OF FEDGKD, FEDGEN, AND PFEDMLKD ON CIFAR-10 AS THE HETEROGENEITY DEGREE ALPHA VARIES

α	FedGen	FedGKD	pFedMLKD
0.2	66.38	61.68	71.59
0.4	71.52	69.81	73.38
0.6	72.59	70.95	75.60
0.8	73.52	73.71	75.69
1.0	74.85	73.67	76.71

TABLE IV
ACCURACY (%) UNDER CLIENT DROPOUT RATES (CIFAR-10, $\alpha = 0.1$)

Dropout Rate	FedAvg	FedProx	FedGKD	pFedMLKD
0%	66.71	59.48	60.35	68.90
10%	64.21	56.82	58.19	67.70
20%	60.83	54.19	54.47	66.12
30%	54.33	49.76	51.01	64.89

during training rounds, mimicking edge device instability. As shown in Table IV, pFedMLKD maintains significantly higher accuracy than baselines like FedAvg and FedGKD across all dropout rates on CIFAR-10 ($\alpha = 0.1$). For example, at a 30% dropout rate, pFedMLKD achieves 64.89% AMP accuracy, outperforming FedAvg (55.23%) and FedGKD (50.01%). This resilience stems from its global ensemble teacher model (12) by aggregating historical client models to reduce reliance on any single update, and multilevel distillation by encoding knowledge hierarchically to withstand partial participation.

To evaluate the scalability and robustness of pFedMLKD with varying client numbers, we conducted experiments on CIFAR-10 under non-IID data partitions ($\alpha = 0.1$) and the number of clients we set to [5, 10, 20, 30, 50]. The results are shown in Fig. 7.

pFedMLKD has demonstrated robust performance in a wide range of client conditions. Its design combines multilevel KD and lightweight model integration, making it ideal for real-world edge FL applications where clients exhibit heterogeneous data and unreliable connections.

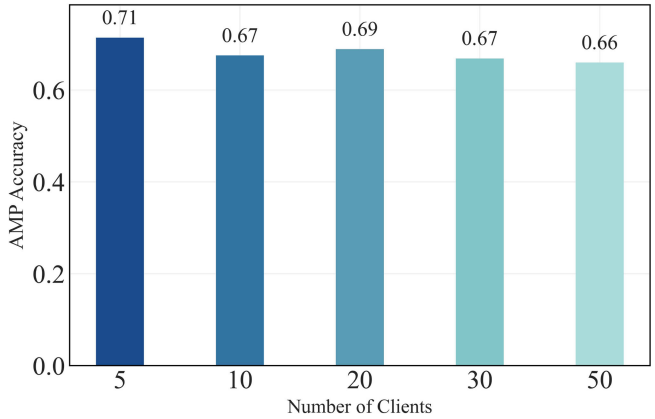
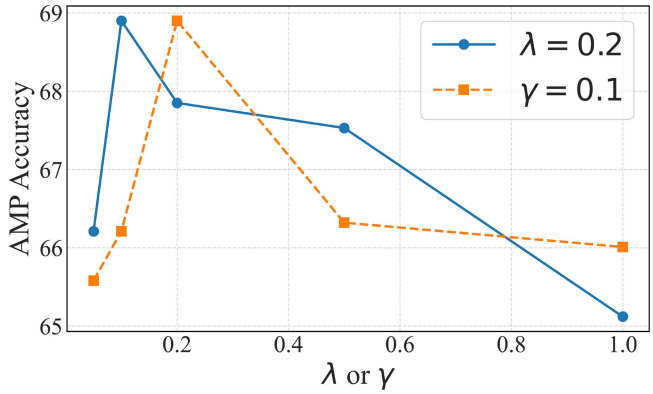
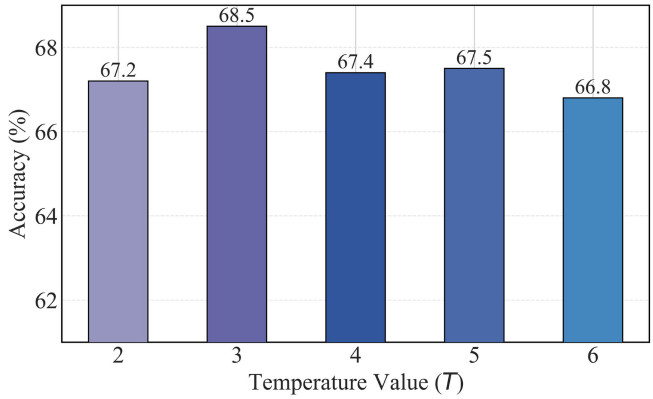


Fig. 7. Model robustness analysis. Taking CIFAR-10 ($\alpha = 0.1$) as an example, we compare the accuracy of the model under the client conditions of [5, 10, 20, 30, 50].



(a)



(b)

Fig. 8. Parameter sensitivity analysis. Experimental results under different hyperparameters. (a) Variation curve of hyperparameters under fixed λ and γ. (b) Experimental results at different temperatures.

D. Parameter Sensitivity Analysis

The distillation coefficients λ (controlling instance-level loss weight) and γ (controlling batch/class-level loss weight) significantly affect model performance. As shown in Fig. 8(a), a

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lower λ undermines instance-level knowledge transfer, causing the model to underfit local data details. Conversely, a higher γ over-regularizes the model, leading to premature convergence to suboptimal solutions. Therefore, we consider $\lambda = 0.2$ and $\gamma = 0.1$ to be a proper selection.

Temperature values regulate the smoothness of logit distributions in KD. In our experiments, we set the temperature to [2, 3, 4, 5, 6] for model enhancement. Next, we conducted temperature sensitivity analysis on CIFAR-10, and the results are shown in Fig. 8(b).

VI. Conclusion and Future Work

In this article, we introduce pFedMLKD, a novel solution designed to address the challenges of data heterogeneity among edge devices in FL. Our proposed KD algorithm innovatively integrates multilevel and global distillation techniques, enabling it to better adapt to the instability of edge devices and effectively address data heterogeneity problem. Moreover, we establish theoretical guarantees for pFedMLKD optimization under nonconvex settings through rigorous convergence analysis. Experimental results demonstrate that our proposed scheme outperforms existing state-of-the-art approaches, delivering substantial performance gains for FL on edge devices while demonstrating its high effectiveness and practical feasibility.

Limitations and Future Work: The notable limitations are as follows. pFedMLKD introduces multiple loss functions on the client side to facilitate the student model to imitate the teacher model, which will generate additional storage overhead. Moreover, the server-side process of integrating the student segment will increase the communication consumption of the algorithm. In future research, we plan to develop effective model storage strategies by analyzing the differences between individual and global models to reduce the storage requirements associated with our methods. In addition, investigating adaptive temperature for the learning model and the teacher model will also be a key focus of research.

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